Module 14

Summarizing Cloud Concepts

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Learning Objectives

- Explain datacenter and storage network architecture.
- Summarize cloud concepts.
- Summarize the use of software, coding, and zero trust in modern network environments.

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Lesson 14.1

Datacenter and Storage Networks

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Datacenter Network Design

Components of Modern Datacenter Network Design

- Virtualization: Agile and scalable networks
- Overlay Networks: Secure, flexible server-to-server traffic
- Zero Trust Architecture: Authentication/authorization for each request
- High Availability features: Networking, power, climate, and access control for reliability

Understanding Traffic Flows: North-South vs. East-West

- North-South Traffic: External client access to datacenter resources
- East-West Traffic: Server-to-server communication within datacenter
- Security and performance challenges: Need for advanced security and efficient traffic management

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Spine and Leaf Topology

Spine and Leaf Topology

- Two-layer network topology for optimized data center traffic flow

Structure

Spine Layer

- Backbone containing top-tier distribution switches
- Spine switches interconnected with all leaf switches but not with each other

Leaf Layer

- Includes access switches that connect to every spine switch in a full mesh topology
- Facilitates direct paths from devices to the network backbone without inter-leaf switch connections

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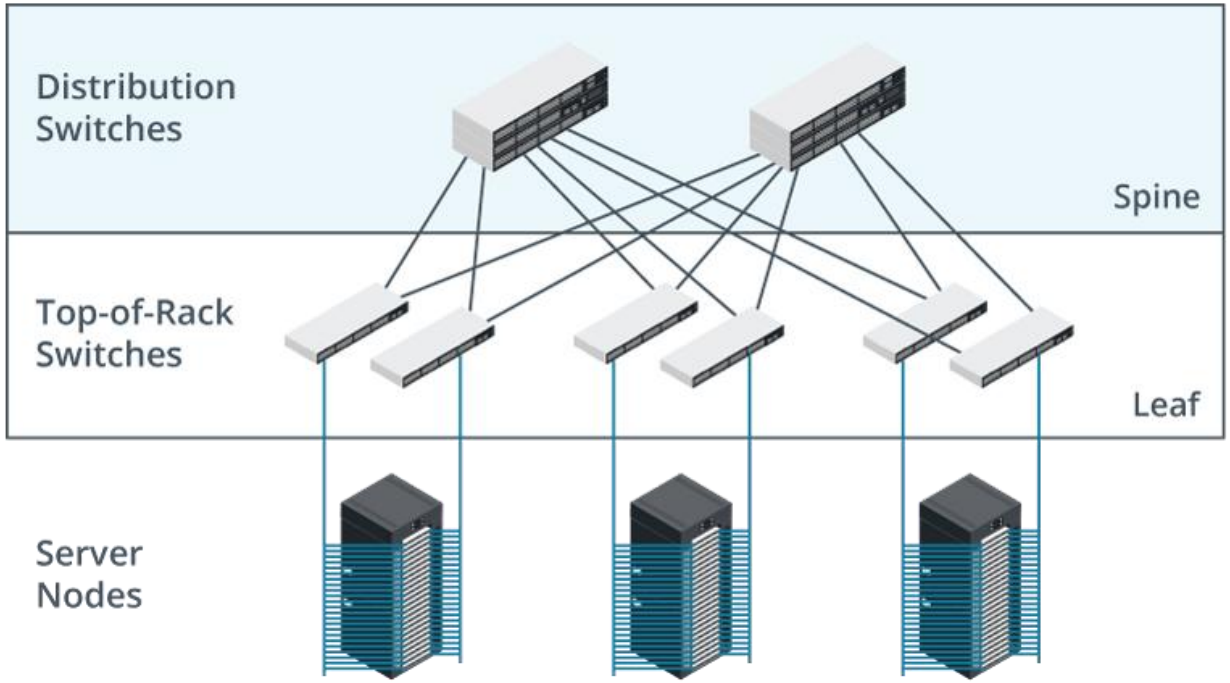
Spine and Leaf Topology Example



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Benefits of Spine and Leaf Topology

Predictable Latency

- Single hop from any server to the backbone
- Ensures consistent network performance

Load Balancing and Failover

- Multiple redundant paths between leaf and spine
- Allow for effective load distribution and reliability

Loop-Free Design

- Non-interconnected spine and use of ECMP on leaf switches
- Eliminates need for Spanning Tree Protocol

Scalability and Redundancy

- Easy to scale by adding more spine or leaf nodes without topology changes
- Servers connect to multiple leaves for redundancy

High-Speed Connectivity

- Top-of-Rack (ToR) switches at the leaf layer provide high bandwidth connections to servers
- Support modern data-intensive applications

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Storage Area Network (SAN)

Storage Area Network (SAN)

- A network dedicated to providing access to consolidated, block-level data storage
- Separates storage resources from the local network

Components of a SAN

- Storage devices (RAID arrays, tape libraries)
- SAN Fabric (Switches and directors that connect servers with storage devices)
- SAN Software (Management and configuration tools)

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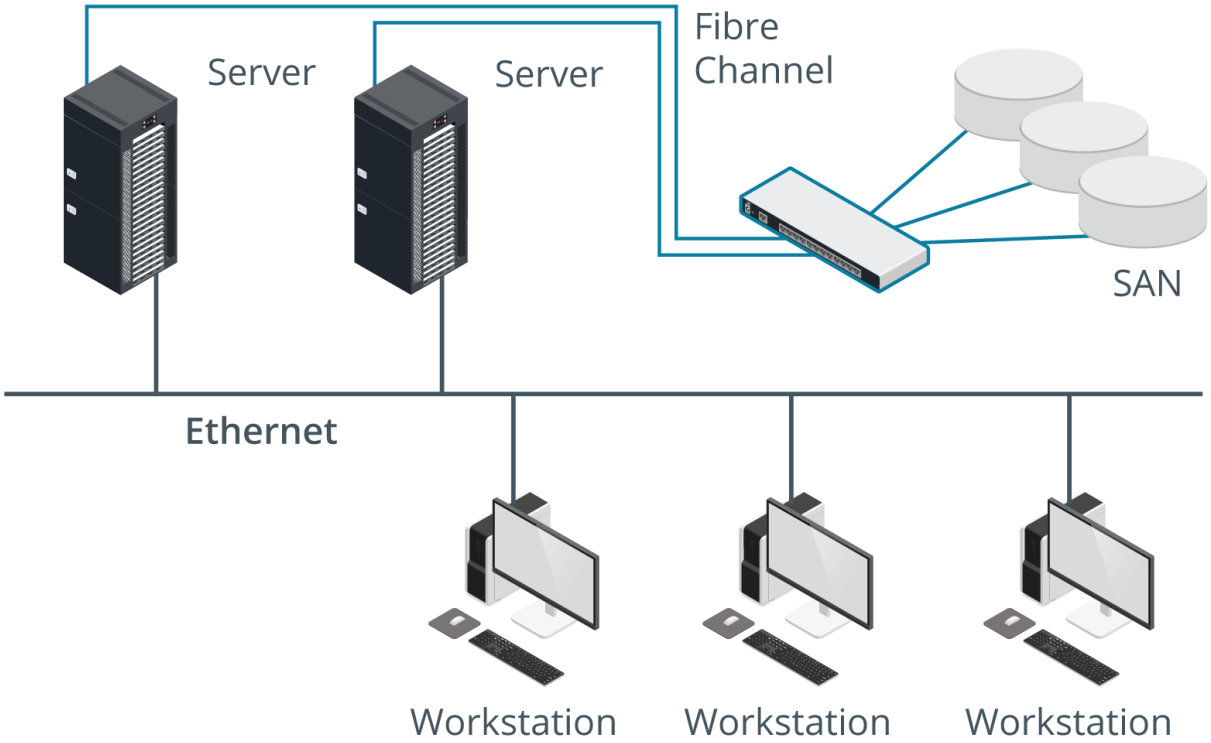
SAN Example



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Storage Area Network (SAN)

Benefits of a SAN

- High Availability
- Scalability
- Minimized latency and maximized throughput

Use Cases

- Database management
- Business continuity
- Cloud storage

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Fibre Channel (FC)

Defined

- High-speed data transfer protocol
- Backbone for SANs
- British spelling “fibre” to distinguish from fiber optic cables

Components

- Initiators (servers with a host bus adapter)
- Targets (storage device ports)
- FC switches (facilitate data transfer)

Advantages

- Performance
- Flexibility
- Speed
- Quality of Service (QoS) mechanisms

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Activity: Pop Quiz

What is the difference between North-South traffic flow and East-West traffic flow?

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Cloud Concepts

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Scalability in Cloud Computing

Definition	Types	Benefits	Examples
• Capability of a cloud computing system to handle growing amount of work • Cloud system grows with your needs: Handles increasing workloads	• Horizontal Scaling (Scaling Out/In) • Vertical Scaling (Scaling Up/Down)	• Flexibility • Performance	• Scale servers for anticipated surge in web traffic (product launch) • Upgrading server specifications to support more intensive compute tasks

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Elasticity in Cloud Computing

Definition	Key Features	Examples
• Ability of a system to automatically adjust and allocate computing resources on demand • Ensures optimal performance despite fluctuations in usage	• Automatic Scaling • Cost-Effectiveness (pay what you use) • Real-Time Responsiveness	• Auto-scaling web application servers during a traffic spike • Decreasing allocated resources overnight when demand is lower to reduce costs

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Cloud Deployment Models

Model	Description	Pros	Cons
Public Cloud	Shared resources over the Internet by CSPs	Cost-effective, pay-as-you-go financing, subscriptions	Higher risks in performance and security
Hosted Private Cloud	Exclusively used by one organization, hosted by a third party	More secure, better performance	More expensive
Private Cloud	Owned and operated by the organization	Greater control over privacy and security	Best for services requiring strict access control
Hybrid Cloud	Mixes public and private cloud solutions	Flexibility to scale; balances security and cost	Complexity in managing multiple environments

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Cloud Service Models

Infrastructure as a Service (IaaS)	Software as a Service (SaaS)	Platform as a Service (PaaS)
• Provides virtualized computing resources over the internet • Examples: AWS EC2, Microsoft Azure Virtual Machines, OpenStack	• Delivers software applications over the internet (subscription basis) • Examples: Microsoft Office 365, Salesforce, Google Workspace	• Hardware and software tools over the internet, typically for application development • Examples: Oracle Database, Microsoft Azure SQL Database, Google App Engine

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Advantages of Cloud Service Models

Scalability and Flexibility

- Scale resources up or down based on demand without the need for physical hardware changes

Cost Efficiency

- Pay only for the computing resources you use

Speed and Efficiency

- Deploy and manage applications faster without the need to set up and maintain physical servers

Accessibility and Collaboration

- Access your applications and data from anywhere

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Content Delivery Networks (CDNs)

What is a CDN?

- Caching solution for high-traffic websites and apps
- Enhances availability, performance, and scalability

How Does It Work?

- Distributes servers and media files near the consumer for faster access
- Content can include web apps, websites, scripts, downloads, and more

Benefits of CDNs

- Distributed across multiple, geographically dispersed data centers
- Quick data access by placing content closer to the consumer
- Easily handles spikes in traffic and data demands

Mechanism

- Content is delivered via multiple Internet backbones, ensuring reliability
- CDN servers replicate content among themselves to stay updated

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Activity: What Would You Do?

Imagine you are the CTO of a growing tech startup that manages sensitive financial data for clients worldwide. You're deciding on the most suitable cloud deployment model to optimize your operations while adhering to strict data privacy laws and regulatory compliance.

Your options include adopting:

- A public cloud for its scalability and cost-effectiveness
- A hosted private cloud for its enhanced security features
- A fully private cloud for ultimate control and privacy
- A hybrid model that combines the benefits of both public and private clouds to meet your specific operational and regulatory needs.

Given the critical nature of the data and the need for scalability as your startup grows, which deployment model would you choose to ensure the best balance of security, compliance, flexibility, and cost?

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Cloud Networking

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Cloud Instances

Types of Cloud Instances

- **Virtual Machines (VMs)**: Fully-fledged servers with emulated hardware, can run an OS and multiple applications
- **Containers**: Isolated environments for running applications (lightweight & portable)
- **Virtual Appliances:** Pre-configured VMs/containers (emulating hardware functions)

Advantages of Cloud Instances

- **Flexibility and Scalability**: Easily resized and scaled; no physical hardware changes
- **Cost-Effectiveness**: Scale costs with resource usage
- **Rapid Deployment**: Fast provisioning; minutes to deploy & configure instances

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Comparing Cloud Instances

Virtual Machines (VMs)

- Run multiple OSs on a single hardware platform
- Complete isolation from the host system (secure and stable operations)
- Higher resource consumption due to emulation of hardware

Virtual Appliances

- Pre-built software packages for tasks like firewalls, ready to run
- Optimized to use resources more efficiently than general-purpose VMs for specific tasks
- Can be easily deployed across diverse cloud environments

Containers

- Shares the host system's kernel, reducing overhead significantly
- Easier to scale and manage due to their small footprint
- Designed to run a single application or service for performance and portability

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Virtual Private Clouds

Virtual Private Clouds (VPCs)

- Serve as isolated environments within a public cloud
- Enable organizations to run their applications securely

Key Features

- Each VPC is isolated from other tenants in the cloud
- Tenants can configure their virtual network
- Offer multiple layers of security controls, including security groups and ACLs

Use Cases

- Application Hosting
- Hybrid Cloud Environments
- Customization and Flexibility
- Networking Across Clouds

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Cloud Gateways

Public vs. Private Subnets

- Public Subnet
  - Requires attaching an Internet gateway and configuring it as the default route
- Private Subnet
  - Remains private without a default Internet gateway route, even if an Internet gateway is attached

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Internet Gateway

Internet Gateways	NAT Gateway	VPN Options	Addressing
• Enables two-way Internet communications • Instances must have a public IP	• Allows outbound Internet access without inbound • No public IP needed for VM	• Securely connect VPCs through software-based VPNs or CSP-managed VPN services	• Public IPs can be from a personal pool or CSP services (Amazon's Elastic IP)

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Cloud Connectivity Options

VPN (Virtual Private Network)	Direct Connect/Colocation	Transit Gateways
• Cost-effective, easy to implement • Best for distributed networks • Challenges: potential latency/bandwidth issues	• High bandwidth, low latency • Ideal for centralized operations • Requires physical infrastructure within or connected to a datacenter	• Simplifies network architecture • Enables multi-VPC and VPC-to-on-premises connections • Offers scalability and ease of management

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Cloud Firewall Security

Cloud Firewall Security

- Acts as a barrier between cloud resources/external threats
- Enforces security rules to allow or block traffic

Functions

- Segmentation for performance and compliance
- Filters traffic based on OSI model layers

OSI Layers Explained

- Layer 3 (Network): IP addresses and TCP/UDP ports filtering
- Layer 4 (Transport): Stateful inspection of connections
- Layer 7 (Application): Parsing protocol headers for content-based filtering

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Implementing Cloud Firewall Solutions

Implementation Options	Considerations	Benefits of Cloud Firewalls	Cost vs. Efficiency
• Host-based software on instances • Service at the virtualization layer	• Software-based solutions consume resources • Management complexity with multiple instances	• Scalable and flexible security management • Facilitates compliance with data protection laws	• Analysis of transaction costs for cloud firewalls • When to consider third-party firewall services

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Security Groups

Definition	Functionality	Key Points
• A security group acts as a virtual firewall for instances to control inbound and outbound traffic	• Operates at the instance level • Provides stateful filtering • Allows specifying allow rules, but all unmatched traffic is dropped	• Multiple security groups can be assigned to a single instance • Customization allows for specific ports and IP ranges to be defined for access • Restricting traffic based on protocol, port numbers, and source/destination IP addresses

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Security Lists

Functionality

- Operates at the subnet level rather than the instance level
- Provides stateless filtering
- Allows setting both allow and deny rules to finely control the traffic flow

Key Points

- A single security list can affect multiple instances within the same subnet
- Ideal for implementing broad network policies that apply uniformly to all instances in a subnet

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Activity: Matching

Fully-fledged servers with emulated hardware, capable of running an operating system and multiple applications

Lightweight, used for running single applications or processes with minimal overhead

Pre-configured; often used for emulating specific hardware functions like routers or firewalls

Containers

Virtual Machines

Virtual Appliances

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Modern Network Environments

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Infrastructure as a Code

Infrastructure as a Code (IaC)

- Management of IT infrastructure through machine-readable definition files
- Uses code to manage and provision the IT infrastructure automatically and safely
- Enables quick, consistent, and cost-effective deployment of applications across various environments

Benefits

- Consistency and efficiency
- Speed and scalability
- Version control and collaboration

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Tools for Infrastructure as a Code

Terraform

- Allows for declarative coding to provision and manage any cloud, infrastructure, or service

Ansible

- Uses a declarative language to automate deployment, configuration, and orchestration tasks

Chef

- Uses recipes and cookbooks to apply configurations across environments

Puppet

- Uses a declarative approach to automate infrastructure management

AWS Cloud Formation

- Allows you to use programming languages or a simple text file to model and provision all the resources needed for your applications

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Use Cases for Infrastructure as a Code

Automating Deployment

- Rapidly deploy VMs and containers to various cloud environments with minimal manual intervention

Consistency & Speed

- Use master images or templates for consistent and fast provisioning of infrastructure

Efficient Upgrades

- Streamline the upgrade process of OS and software versions

Dynamic Inventory Management

- Keep track of cloud instances through dynamic inventories

Reusable Tasks & Scalability

- Create modular, reusable blocks of code for various tasks

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Source Control

Source Control

- Method to manage and track code changes in software development
- Essential for collaborative projects and maintaining code integrity

Components

- Version control
- Central repository
- Branching
- Conflict identification

Benefits of Source Control

- Enhances collaboration among developers
- Facilitates efficient management of code changes
- Maintaining a history of project changes and revisions

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Software-Defined Networking (SDN)

Definition

- Simplifies network management by separating control and forwarding functions
- Allows direct programmability of network control and abstraction of the infrastructure for applications and services

Components

- SDN Controller (Brain of the network)
- Southbound APIs (Communication between controller and devices)
- Northbound APIs (Interface between controller and the business applications)

Benefits

- Enhanced network agility and flexibility
- Simplified network management
- Improved network efficiency and performance

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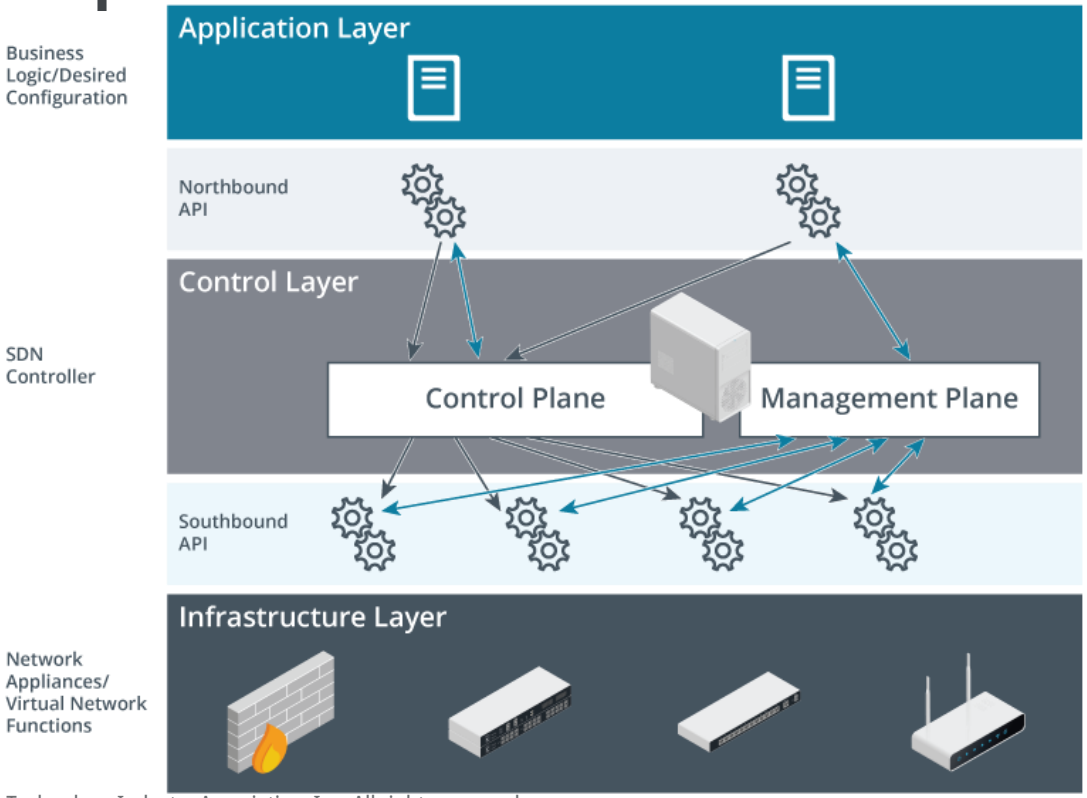
SDN Example



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SDN Use Cases

Network Virtualization

- Creating multiple virtual networks with distinct security policies and functions over a single physical infrastructure

Cloud Computing

- Dynamically adjusting network resources to meet the demands of cloud services, improving scalability and reducing operational costs

Data Center Management

- Automated provisioning and management of network resources, enhancing data flow efficiency and reducing latency

IoT (Internet of Things):

- Managing the massive and diverse traffic from IoT devices for reliable connectivity and security at scale

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Overlay Networks

Overlay Network

- Virtual network built on top of an existing physical network
- Allows for logical links between nodes or different networks

Examples

- Virtual Private Networks (VPNs)
- Networks using encapsulation protocols SDN

Components

- Uses encapsulation to create virtual tunnels
- Virtual Extensible LAN (VXLAN) for data center implementations
- Facilitates network segmentation and isolation

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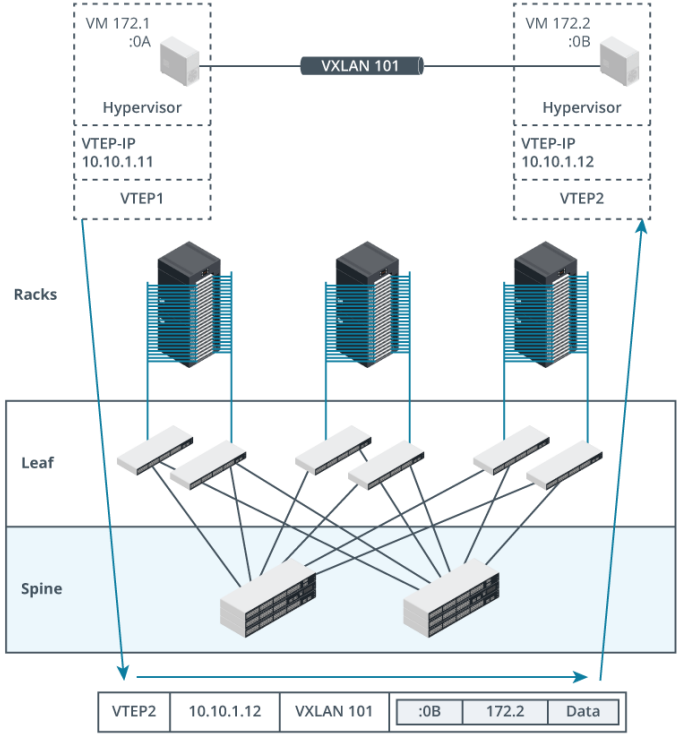
Overlay Network Example



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Software-Defined WAN

Software-Defined WAN	Components	Benefits
• Innovative approach to designing and deploying enterprise WAN • Enhanced connectivity between branch offices, datacenters, and the cloud • Traffic flow optimization: automation, central management, and secure IPsec tunnels	• SD-WAN Controller & Management Software • SD-WAN Devices	• Improved Performance • Increased Security • Cost Savings

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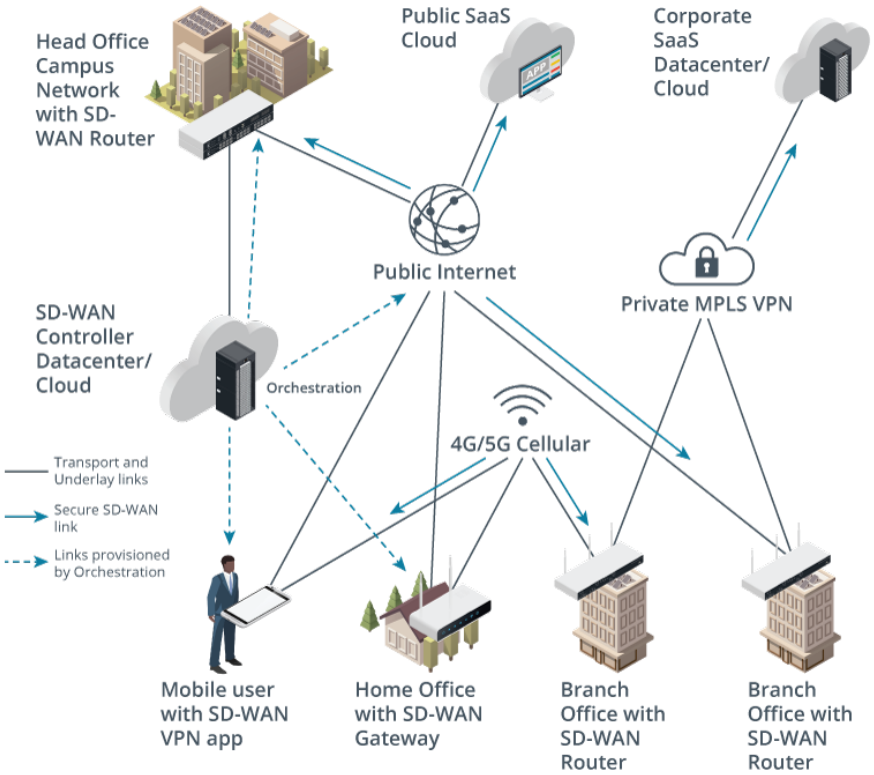
Software-Defined WAN Example



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Zero Trust Architecture – Core Principles

Assume No Trust

Operate under the assumption that no entity (user or device) is trusted by default (inside/ outside network)

Continuous Verification

Implement constant authentication and authorization of every user/device for every session/ transaction

Least Privilege Access

Grant users and devices minimum levels of access (permissions) needed to accomplish tasks to minimize potential attack vectors

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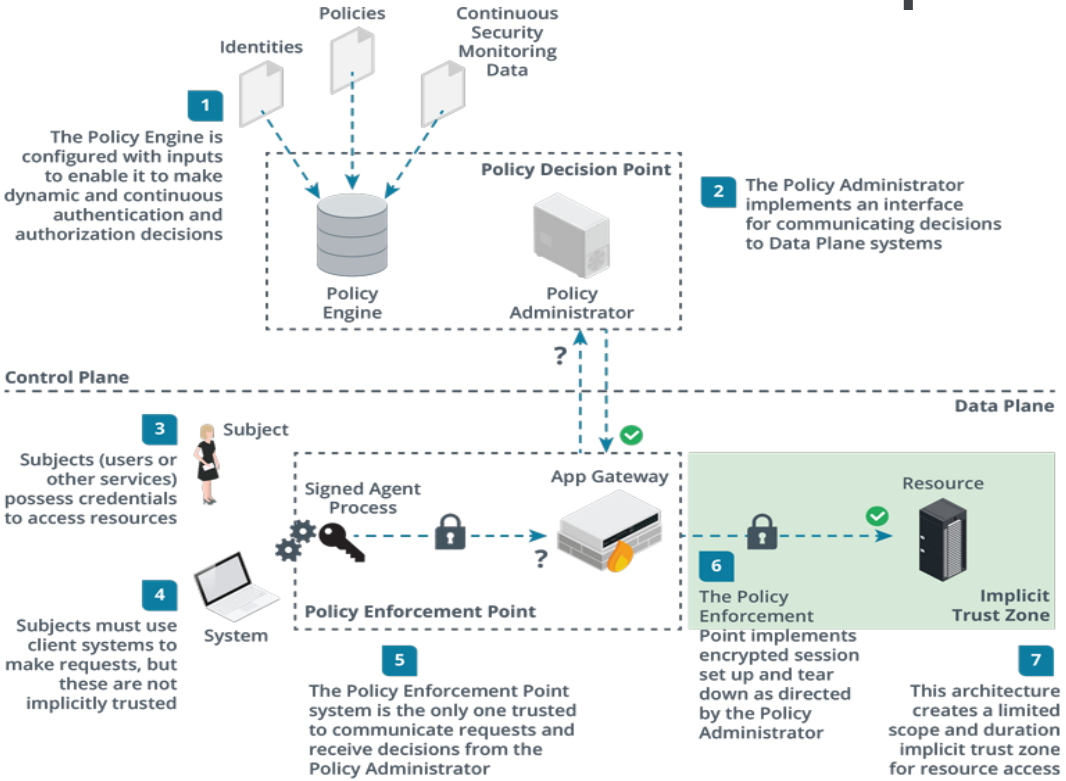
Zero Trust Architecture Example



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Implementing Zero Trust Architecture

Policy-driven Authentication and Authorization

Use centralized policy decision points for dynamic security assessments based on user behavior, device health, and network context

Separation of Control and Data Planes

Maintain distinct mechanisms for policy definition and enforcement to reduce risks and ensure granular access control

Implicit Trust Zones and Microsegmentation

Establish secure pathways for data based on stringent policy enforcement

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Secure Access Service Edge (SASE)

Secure Access Service Edge (SASE)

- Cloud-based architecture combining network and security functions with widespread global reach
- Integrates SD-WAN technology and Security Service Edge (SSE)

Benefits of SASE

- Simplifies WAN architecture
- Enhances security posture
- Reduces complexity and overhead
- Improves performance and user experience

Components

- SD-WAN Technology
- Secure Service Edge (SSE)

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Secure Service Edge (SSE)

Functions of SSE

- Zero Trust Architecture
- Cloud Access Security Broker (CASB)

Functions of CASB

- Manages and enforces security policies
- Scans for malware
- Monitors and audits activity
- Mitigates data exfiltration

CASB Implementation Methods

- Forward Proxy: Acts as a gatekeeper, monitoring and controlling outbound client traffic
- Reverse Proxy: Cloud gatekeeper, enforces access policies for services
- API: Direct connections for cloud service access & control

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Summary

- **Cloud Strategy**: Evaluate cloud apps & services; choose deployment model (public, private, hybrid) & service type (IaaS, SaaS, PaaS, DaaS) for optimal scalability & elasticity.
- **Cloud App Development**: Leverage IaC for automation and leverage SDN benefits.
- **Private Cloud/Datacenter**: Virtualization & SAN for enhanced elasticity & scalability. Integrate with SDN & NFV.
- **Public Cloud Security**: Define Cloud Responsibility Matrix and conduct regular risk assessments & security audits.

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